

Divergence of the Irreducible Polarizability $\Pi(q)$ in the 3D Homogeneous Electron Gas at $r_s \gtrsim 5.25$: Compressibility Instability without Coulomb

Analysis Notes
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I. SETUP AND DEFINITIONS

We consider the three-dimensional homogeneous electron gas (HEG) at density parameter r_s , with Fermi wavevector $k_F = (9\pi/4)^{1/3}/r_s$, density $n_0 = 3/(4\pi r_s^3)$, and density of states at the Fermi level $N_F = k_F/\pi^2$.

Three static density response functions are compared:

1. Lindhard function (non-interacting):

$$\chi_0(q) = -\frac{k_F}{2\pi^2} \left[1 - \left(\frac{Q}{4} - \frac{1}{Q} \right) \ln \left| \frac{Q+2}{Q-2} \right| \right], \quad (1)$$

where $Q = q/k_F$. This is entirely analytical and well-behaved.

2. Full (reducible) response function:

$$\chi(q) = \frac{\chi_0(q)}{1 - \chi_0(q)[v_c(q) + f_{xc}(q)]}, \quad (2)$$

where $v_c(q) = 4\pi/q^2$ is the bare Coulomb interaction and $f_{xc}(q)$ is the static exchange–correlation kernel parametrized via the Corradini–Ortiz–Perdew–Zieschang (COPZ) local field factor [1]:

$$f_{xc}(q) = -v_c(q) G(q; r_s). \quad (3)$$

3. Irreducible polarizability:

$$\Pi(q) = \frac{\chi_0(q)}{1 - \chi_0(q) f_{xc}(q)}. \quad (4)$$

This is linked to the full response via the exact relation $\chi(q) = \Pi(q)/[1 - v_c(q)\Pi(q)]$.

II. THE OBSERVED DIVERGENCE

When evaluating these expressions numerically for increasing r_s , $\chi_0(q)$ and $\chi(q)$ remain smooth and finite for all r_s . However, $\Pi(q)$ develops a divergence for $r_s \gtrsim 5.25$.

The divergence occurs when the denominator of Eq. (4) vanishes:

$$1 - \chi_0(q) f_{xc}(q) = 0 \quad \text{for some } q. \quad (5)$$

Numerical evaluation reveals that this condition is first satisfied at $q \rightarrow 0$ (not near $2k_F$), at a critical density parameter

$r_s^{(c)} \approx 5.24$

 (Corradini–PZ parametrization). (6)

Table I summarizes the minimum value of the denominator $D(q) \equiv 1 - \chi_0(q) f_{xc}(q)$ across all q , for several values of r_s .

TABLE I. Minimum of the Dyson denominator for $\Pi(q)$ and $\chi(q)$ as a function of r_s . The Π -denominator crosses zero near $r_s \approx 5.24$, while the χ -denominator remains safely positive.

r_s	$\min[1 - \chi_0 f_{xc}]$	$q_{\min}/k_F$	$\min[1 - \chi_0(v_c + f_{xc})]$
1	+0.8129	0.82	+0.9994
2	+0.6369	0.70	+0.9924
3	+0.4514	0.47	+0.9804
4	+0.2543	0.02	+0.9682
5	+0.0494	0.03	+0.9563
5.24	≈ 0	$\rightarrow 0$	+0.9539
6	−0.1605	0.03	+0.9447
8	−0.5928	0.04	+0.9227
10	−1.0386	0.05	+0.9020

III. PHYSICAL ORIGIN: COMPRESSIBILITY INSTABILITY

The divergence is *not* a numerical artifact. It has a clear physical interpretation rooted in the **compressibility sum rule**.

A. The $q \rightarrow 0$ limit

In the long-wavelength limit, the Lindhard function approaches

$$\chi_0(q \rightarrow 0) = -\frac{k_F}{2\pi^2} \equiv -N_F, \quad (7)$$

and $f_{xc}(q \rightarrow 0) = f_{xc}(0)$ approaches a finite, *negative* constant (dominated by the attractive exchange hole). Consequently, the product

$$\chi_0(0) f_{xc}(0) = \underbrace{(-N_F)}_{\text{negative}} \times \underbrace{f_{xc}(0)}_{\text{negative}} = +|N_F f_{xc}(0)| > 0. \quad (8)$$

This product grows with r_s because:

- $|f_{xc}(0)|$ grows rapidly (exchange–correlation effects strengthen at low density), scaling roughly as $\sim r_s^2$;
- $N_F \propto 1/r_s$ decreases, but more slowly than $|f_{xc}|$ increases.

Very explicitly, $1/\chi_0(0) \propto r_s$ grows linearly, but $|f_{xc}(0)|$ grows faster. The crossover occurs when they balance:

$$N_F |f_{xc}(0)| = 1 \iff r_s = r_s^{(c)} \approx 5.24. \quad (9)$$

The numerical values confirm this:

r_s	$-1/\chi_0(0)$	$f_{xc}(0)$	$1-\chi_0 f_{xc}$
1	5.14	-0.88	+0.83
3	15.43	-8.43	+0.45
5	25.71	-24.44	+0.05
5.24	26.95	-26.96	≈ 0
6	30.86	-35.81	-0.16
10	51.43	-104.86	-1.04

B. Interpretation as a compressibility divergence

The $q = 0$ response is directly related to the thermodynamic compressibility. For the full interacting system:

$$\chi(q=0) = -n_0^2 \kappa, \quad (10)$$

where κ is the isothermal compressibility and the inverse compressibility receives contributions:

$$\frac{1}{n_0^2 \kappa} = \frac{1}{n_0^2 \kappa_0} - f_{xc}(0) - v_c(0). \quad (11)$$

The Coulomb term $v_c(0) = 4\pi/q^2 \rightarrow +\infty$ as $q \rightarrow 0$, which always ensures $\kappa > 0$ (the physical system is stable against uniform compression).

For $\Pi(q)$, however, Coulomb is excluded by construction. The “effective compressibility without Coulomb” is

$$\frac{1}{n_0^2 \kappa_{xc}} = \frac{1}{n_0^2 \kappa_0} - f_{xc}(0) = \underbrace{\frac{2E_F}{3n_0}}_{\text{kinetic pressure}} + \underbrace{|f_{xc}(0)|}_{\text{xc “attraction”}}, \quad (12)$$

where the sign of $f_{xc}(0)$ makes the xc contribution reduce (and eventually reverse) the kinetic pressure. The divergence $\kappa_{xc} \rightarrow \infty$ signals that **exchange–correlation attraction alone overcomes the kinetic (Pauli) pressure**, producing an instability in the hypothetical system with f_{xc} but no Coulomb repulsion.

IV. WHY $\chi(q)$ REMAINS STABLE

The full response $\chi(q)$ includes Coulomb in its Dyson equation:

$$1 - \chi_0(q)[v_c(q) + f_{xc}(q)]. \quad (13)$$

At small q , $v_c(q) = 4\pi/q^2 \gg |f_{xc}(q)|$, ensuring the denominator is dominated by

$$1 - \chi_0(q) v_c(q) \approx 1 + \frac{4\pi N_F}{q^2} \rightarrow +\infty \quad (q \rightarrow 0). \quad (14)$$

This is the standard screening result: the long-range Coulomb repulsion ensures that the charge response is always finite and well-behaved, regardless of how strong f_{xc} becomes. In the language of dielectric screening, the dielectric function $\varepsilon(q) \rightarrow 1 + k_{TF}^2/q^2 \rightarrow \infty$ as $q \rightarrow 0$, so $\chi(q \rightarrow 0) = \chi_0(q)/\varepsilon(q) \rightarrow 0$.

The physical electron gas remains stable for all metallic densities (Wigner crystallization occurs only at $r_s \gtrsim 100$).

V. IS THIS PHYSICAL OR AN ARTIFACT?

The answer is **both**—it reflects real physics but may be exaggerated by the parametrization.

A. Genuine physics: negative compressibility

The electron gas is known to have a *negative* contribution to the compressibility from exchange–correlation at low densities. The total energy per particle is

$$\varepsilon(r_s) = \frac{\varepsilon_{\text{kin}}}{r_s^2} + \frac{\varepsilon_x}{r_s} + \varepsilon_c(r_s), \quad (15)$$

with $\varepsilon_{\text{kin}} = 2.21$ Ry, $\varepsilon_x = -0.916$ Ry (in Rydbergs). The compressibility $\kappa^{-1} \propto r_s^2 d^2(r_s \varepsilon)/dr_s^2$ goes through zero at some r_s when only kinetic + xc are included. The value of $r_s^{(c)}$ depends on the correlation parametrization; the Perdew–Zunger (PZ) form used in Corradini’s COPZ gives $r_s^{(c)} \approx 5.24$.

This is *not* a phase transition of the real electron gas (Coulomb always prevents it), but it does mean that $\Pi(q)$ computed via the Dyson equation with a static f_{xc} becomes *ill-defined* beyond this r_s .

B. Subtleties of the Corradini parametrization

The COPZ local field factor $G(q; r_s)$ is constructed to satisfy:

- The **compressibility sum rule** at $q = 0$: $G(0) = -q^2 f_{xc}(0)/(4\pi)$, which is built in via the PZ correlation energy parametrization and its derivatives.
- The **third-frequency-moment sum rule** at $q \rightarrow \infty$: $G(q \rightarrow \infty) \rightarrow B + Cq^2 + \dots$.
- Smooth interpolation between these limits using QMC data from Moroni *et al.*

The divergence of $\Pi(q)$ is thus a *direct consequence* of the PZ correlation energy having a large enough $|d\mu_c/dn|$ to satisfy the instability condition. Different correlation parametrizations (VWN, PW92) would shift $r_s^{(c)}$ slightly but not eliminate the phenomenon.

C. The two routes to $\Pi(q)$ are algebraically identical

One might hope to circumvent the divergence by computing $\Pi(q)$ from the well-behaved $\chi(q)$ via

$$\Pi(q) = \frac{\chi(q)}{1 + v_c(q) \chi(q)}. \quad (16)$$

However, substituting $\chi = \chi_0/[1 - \chi_0(v_c + f_{xc})]$:

$$\begin{aligned} \Pi &= \frac{\chi_0/[1 - \chi_0(v_c + f_{xc})]}{1 + v_c \chi_0/[1 - \chi_0(v_c + f_{xc})]} \\ &= \frac{\chi_0}{1 - \chi_0(v_c + f_{xc}) + v_c \chi_0} \\ &= \frac{\chi_0}{1 - \chi_0 f_{xc}}. \end{aligned} \quad (17)$$

The Coulomb potential cancels *exactly*. The two routes are *algebraically identical*—both produce precisely $\Pi = \chi_0/(1 - \chi_0 f_{xc})$.

Therefore, the divergence of $\Pi(q)$ when $1 - \chi_0 f_{xc} = 0$ is **intrinsic to the definition of the irreducible polarizability** given a particular f_{xc} . There is no algebraic rearrangement that avoids it: the pole in Π is a genuine feature of the Corradini–PZ kernel at $r_s > r_s^{(c)}$.

D. Why doesn't QMC $\Pi(q)$ diverge?

Direct QMC simulations of the density response measure the *full* $\chi(q)$, from which one can extract $\Pi(q)$ via Eq. (16). This “exact” $\Pi(q)$ does not diverge because the QMC structure factors implicitly contain frequency-dependent (dynamic) xc effects that the static COPZ f_{xc} cannot capture.

The divergence arises because a *static, analytic* parametrization of $f_{xc}(q)$ is being used beyond its physical range of validity. The Corradini–PZ form enforces the compressibility sum rule $f_{xc}(0) = d\mu_{xc}/dn$, which faithfully reproduces the large (negative) xc compressibility contribution at low density. This is correct for $\chi(q)$, where v_c always dominates, but it causes $\Pi(q)$ to diverge because Π probes the system without Coulomb stabilization.

VI. PRACTICAL CONSEQUENCES

1. **For $r_s < 5.2$:** All three functions (χ_0 , Π , χ) are well-behaved and can be compared directly.
2. **For $r_s > 5.25$:** $\Pi(q)$ is divergent *regardless* of how it is computed—whether directly via $\chi_0/(1 - \chi_0 f_{xc})$ or indirectly via $\chi/(1 + v_c \chi)$ —because both routes are algebraically identical (Sec. V C). There is no workaround within the static COPZ kernel.
3. **Fourier transforms:** Since $\Pi(q)$ diverges at small q , its Fourier transform $\Pi(r)$ will also exhibit unphysical large-amplitude oscillations or blow up entirely.
4. **Fitting and interpolation:** Any fitting procedure (e.g., fitting $\Delta\Pi(r) = \Pi(r) - \chi_0(r)$ with damped oscillatory models) will fail for $r_s > r_s^{(c)}$ if using the static COPZ f_{xc} .

VII. RECOMMENDATION

For the interpolation workflow at arbitrary r_s :

1. **Work with χ , not Π :** Compute $\chi(q)$ from the well-behaved Dyson equation (2), which never diverges. Perform all fitting and interpolation on $\Delta\chi(r) = \chi_0(r) - \chi(r)$ rather than $\Delta\Pi(r)$.
2. **Do not use $\Pi(q)$ for $r_s > 5$:** Both computational routes to $\Pi(q)$ are equivalent (Sec. V C) and both diverge. The divergence is an intrinsic consequence of the static COPZ f_{xc} and cannot be circumvented by algebraic rearrangement.
3. **If Π is essential:** Use a frequency-dependent $f_{xc}(q, \omega)$ or extract Π directly from QMC structure factor data, bypassing the static Dyson equation. Alternatively, use a different f_{xc} parametrization (e.g., one that regularizes the $q \rightarrow 0$ divergence), keeping in mind that the critical $r_s^{(c)}$ will shift but not disappear.

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 [4] G. F. Giuliani and G. Vignale, *Quantum Theory of the Electron Liquid* (Cambridge University Press, 2005).